



The *Cyclopaedia* summarized human knowledge, reflecting the growing belief that people could learn about the world by studying it scientifically.

IN 18TH CENTURY INDIA there was no better symbol of power and enlightenment than knowledge of the heavens, which may be why the **Maharajah of Amber Jai Singh II** had five massive observatories built across his kingdom. The greatest of them was the Jantar Mantar at Jaipur, begun in 1727, which still stands today. **Jantar Mantar means “calculation instrument”** and this site contains the world’s largest sundial, the Samrat Yantra, which is accurate to within two seconds. Its significance is as much astrological and religious as it is scientific.

In the same year, English clergyman and naturalist **Stephen Hales** wrote about his experiments on plant physiology in *Vegetable Staticks*. He noticed how plants drew water up through their stems due to **root pressure and transpiration** (the evaporation of water through the leaves). He also suggested that plants absorb food from air using energy from sunlight—an



Jaipur’s Jantar Mantar

The Samrat Yantra in the Jantar Mantar is the world’s biggest sundial, at over 88 ft (27 m), and the Sun’s shadow can be seen visibly moving over 1/2 in every 10 seconds.

idea eventually leading to our understanding of **photosynthesis** (see 1787–88).

In 1728, English physicist **James Bradley** looked at the stars to make one of the first **accurate measurements of the speed of light**. He used **stellar aberration**, the apparent movement of stars caused by Earth’s motion, which he had discovered in 1722. Bradley measured the stellar aberration of starlight from a star in the constellation of Draco and

calculated the speed of light to be 987,532,800 ft/s (301,000,000 m/s), remarkably close to today’s estimate of 983,571,056 ft/s (299,792,458 m/s).

In Paris, French physician **Pierre Fauchard** launched **modern dentistry** in his book *Le Chirurgien Dentiste* (*The Surgeon Dentist*). He introduced fillings and advocated cutting down on sugar to avoid tooth decay. In London, English writer **Ephraim Chambers** published *The Cyclopaedia* or *A Universal Dictionary of Arts and Sciences*, one of the first great **encyclopedias of knowledge** written in English.

90 FEET
THE HEIGHT OF THE
SAMRAT YANTRA
SUNDIAL NEEDLE
AT JANTAR MANTAR

“...AS THE TUBE COMMUNICATED A **LIGHT TO BODIES...** MIGHT [IT] NOT AT THE SAME TIME **COMMUNICATE ELECTRICITY TO THEM...**”

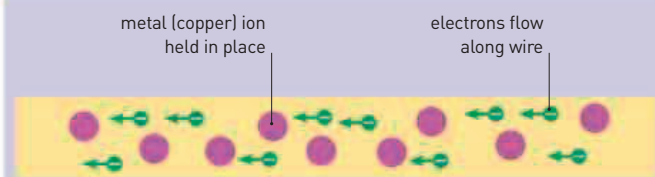
Stephen Gray, English experimenter, in *A Letter to Cromwell Mortimer Containing Several Experiments Concerning Electricity*, 1731

FOR MUCH OF HIS LIFE, ENGLISHMAN STEPHEN GRAY worked in the family trade as a dyer, and appears to have been largely self-educated. When he retired in the 1720s he began experimenting with electrical effects. It was the simplicity of Gray’s experiments that introduced many people to the phenomenon of electricity. Most significantly, he demonstrated how an **electric charge** could be **transmitted over distances** by showing that it could be conducted through a damp silk thread for hundreds of yards.

In France, the prodigious mathematician and astronomer **Pierre Bouguer** was making key discoveries about the transmission of light. Appointed a professor and lecturer in physics

and mathematics when he was just 15 years of age, Bouguer began to study how **light is absorbed by transparent substances** such as the atmosphere. He found that light does not decrease in intensity arithmetically (uniformly) as it passes through the air but geometrically (at an ever-increasing rate).

It is not just the atmosphere that distorts starlight. **Telescopes** of the day suffered from **chromatic aberration**—the blurring and color fringing caused by the fact that a simple, conventional lens cannot focus all the different wavelengths of light at the same point. British inventor **Chester Moor Hall** solved this problem by producing the first **achromatic lenses**.



ELECTRICAL CONDUCTION

Electrical conduction is the movement of electrical charge. It is essentially a relay race of electrons (discovered later, in 1897). Electrons are normally attached to atoms, but can sometimes break free. The more easily electrons can break free, the better a substance can conduct electricity, which is why metals such as copper are good conductors.

